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Static Antisolvent Processing for Scalable and Sustainable Perovskite Photovoltaics

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Abstract

Metal halide perovskite solar cells have achieved remarkable laboratory-scale efficiencies, but their commercialization requires processing strategies that are reproducible, scalable, sustainable, and broadly applicable. Here, we introduce a static antisolvent treatment applied after film deposition. By incorporating isopropanol into a conventional low-polarity antisolvent, we accelerate and homogenize perovskite crystallization through enhanced polarity and improved interaction with precursors and deposition solvents. In situ grazing-incidence wide-angle X-ray scattering and photoluminescence reveal the formation of larger, more uniformly distributed crystalline nanodomains during static treatment with the binary antisolvent. The method is effective for four perovskite compositions under 30% relative humidity, and replacing aromatic antisolvents with eco-friendly ethyl acetate preserves film quality while improving sustainability. We further validate the approach in 25 cm² single-junction minimodules and perovskite/silicon tandem devices, achieving power conversion efficiencies of 18.1% and 27.7%, respectively. Taken together, these findings provide a scalable and sustainable route for the fabrication of perovskite photovoltaics.

Introduction

Metal halide perovskites have rapidly emerged as prominent materials for photovoltaic technologies owing to their low-impact solution-based fabrication and record power conversion efficiencies approaching 27%.^{1–4} The rapid progress achieved by perovskite-based devices has been largely driven by solvent engineering strategies.^{5–7} In particular, the antisolvent-assisted crystallization step widely used to control perovskite film formation relies on dynamic solvent-dripping processes that are highly sensitive to processing conditions and challenging to reproduce over large areas.^{8–14} Previous studies have explored post-deposition solvent treatments using single-solvent baths, such as toluene, diethyl ether, ethyl acetate or chloroform.^{15–21} Jang et al. reported a cold diethyl ether antisolvent bath, in which the perovskite wet film was submerged at 0 °C to retard nucleation and growth kinetics, thereby enabling the formation of a (110)-oriented perovskite layer.²² A similar strategy has been employed to obtain a smooth and dense wide-bandgap mixed-halide perovskite by enhancing the nucleation rate of a bromide-rich perovskite composition.¹⁶ Recently, Zhang et al. developed an antisolvent bathing strategy that uses alkyl chloride solvents, such as chloroform, as the bathing agent to process the perovskite layer under ambient conditions, achieving efficiencies above 24%.²⁰

Although promising, these approaches have mainly been demonstrated in small-area single-junction devices and for single perovskite compositions. Moreover, the need for strict control of inert-atmosphere processing conditions, together with the toxic nature of commonly used solvents, often aromatic or halogenated, still hinders the development of robust and environmentally compatible fabrication routes for industrial-scale perovskite photovoltaics.^{23–25} Developing scalable and eco-compatible crystallization strategies that ensure uniform film formation across different perovskite compositions and device architectures, therefore, remains a critical challenge. In this context, we propose a strategy to trigger perovskite crystallization through post-deposition solvent manipulation, using an ambient air static antisolvent treatment. Our method exploits a mixture of a low-polarity solvent (chlorobenzene or eco-

friendly ethyl acetate) and a polar solvent (isopropanol) to promote a controlled crystallization of the perovskite film. Using advanced in situ Grazing-Incidence Wide-Angle X-ray Scattering (GIWAXS) and in situ photoluminescence (PL), we investigate the crystallization dynamics of the perovskite thin films during their interaction with single or mixed solvent systems. Regardless of the specific low-polar solvent employed, we find that the addition of a polar component is essential to accelerate and homogenize the perovskite crystallization, to larger crystalline domains. Even though only a small amount of IPA is present, its addition slightly increases the polarity of the antisolvent compared to pristine CB and modifies its diffusivity within the perovskite wet film.²⁶ Furthermore, isopropanol may promote the redistribution of organic components that are soluble in IPA but not in CB.^{27,28} Overall, the combination of these two effects leads to improved crystalline homogeneity and enhanced reproducibility of photovoltaic performance, outcomes that are not achievable using a single low-polar solvent. Notably, the mixed-solvent treatment promotes stabilization of the cubic phase even when films are processed under ambient humidity conditions (relative humidity ~30%), while maintaining high reproducibility. We further demonstrate the generality of this approach across four perovskite compositions spanning bandgaps from 1.53 to 1.68 eV. Importantly, tuning the polarity of the solvent mixture enables the incorporation of dopants and additives that would otherwise be insoluble in low-polarity solvents alone, thereby reducing the number of fabrication steps by combining perovskite crystallization and passivation into a single process.²⁹ Finally, we validate the scalability of this strategy in both perovskite/silicon tandem solar cells and single-junction minimodules with an active area of 25 cm², achieving power conversion efficiencies of 27.7% and 18.1%, respectively. Taken together, these results demonstrate the robustness, scalability, and environmental compatibility of the process across large-area and tandem architectures, establishing it as a viable and industrially relevant route toward perovskite photovoltaic manufacturing.

Results and discussion

Protocol design and *in situ* analysis

In order to develop an antisolvent method compatible with large-scale deposition techniques, we replaced the dynamic antisolvent approach, with a static antisolvent method. **Figure 1a** illustrates the static antisolvent procedure. After depositing the formamidinium tri-iodide (FAPbI₃) perovskite layer, the film is immersed in the antisolvent bath. It is widely accepted that the solvent's chemical-physical properties can have a strong impact on perovskite crystallization.^{30–32} For this reason, we explored the effect of different antisolvent systems with varying polarity, obtained by introducing different amounts of isopropanol (IPA), a solvent commonly used in perovskite processing, as a second component in chlorobenzene (CB), on the crystallization of FAPbI₃.^{33–35} Due to its low polarity, IPA ensures a negligible damage of the perovskite layer during the antisolvent treatment.^{36–38} X-ray diffraction (XRD) patterns of FAPbI₃ layers treated with pure CB and different CB/IPA ratios are reported in **Figures 1b** and **S1**. By analyzing the full width at half maximum (FWHM) of the (100) reflection of the cubic α -phase of FAPbI₃ at $2\theta = 13.9^\circ$, and considering its inverse relationship with crystallite size according to the Scherrer equation, the largest crystallite size was observed for sample the sample treated with a 12:1 v/v CB/IPA mixture (named hereafter CB/IPAs12).³⁹ Besides, we tested the impact of the soaking time on the crystal quality and phase homogeneity using XRD and PL mapping measurements. The results obtained from both characterization techniques consistently indicate a significant improvement in the film properties during the first 10 seconds of the dipping process. Specifically, the XRD data (**Figure S2**) reveal enhanced crystallinity and a more pronounced development of the perovskite phase (as shown by the decreasing trend of the $\frac{\text{FWHM}}{\text{Peak intensity}_{(100)}}$ ratio), while the PL imaging measurements (**Figure S3**) demonstrate increased spatial uniformity and PL signal intensity after the first 10 seconds of treatment, reaching a maximum at 20 seconds of treatment.

To validate the quality of the resulting perovskite layer, we integrated the optimized solvent mixture into the fabrication of p–i–n single-junction devices. The control device was prepared using the conventional dynamic antisolvent dripping method with CB only (**Figure S4**).⁴⁰ Distribution reported in **Figures S5** and **1c** corroborates the trend observed in the XRD analysis. Using the static CB/IPAs12 mixture higher and more reproducible PV performance is obtained with a net improvement with respect to the devices treated with static CB (CBs), achieving power conversion efficiencies (PCE) around 21% that are comparable to the control device (dynamic method).

To understand the beneficial impact of IPA addition in the static approach, we investigated CBs and CB/IPAs12 samples by in-situ photoluminescence (PL) during the soaking step. The results are shown in **Figures 1d** and **1e**. Comparing the two samples, two main differences emerge. First, the CBs samples exhibit a lower PL intensity than the CB/IPAs12 sample. Second, the CBs emission after the antisolvents soaking is broader and blue-shifted with respect to CB/IPAs12. Since the samples were not annealed, the signal can be attributed to nanodomains of perovskite cubic-phase. The broader and blue-shifted emission of CBs indicates the formation of particles with a smaller and more inhomogeneous size distribution compared to CB/IPAs 12. Moreover, the redshift of the CB/IPAs 12 signal toward lower energies, closer to the bandgap of FAPbI₃, suggests that this mixture promotes α -phase crystallization already at room temperature. By analysing the PL spectra and PL maps collected from different regions of the CBs and CB/IPAs12 samples, we observed higher PL intensity and greater spatial homogeneity in the CB/IPAs12 sample. Specifically, the PL signal variation remained within 15% of the maximum intensity for CB/IPAs12, whereas it reached up to 23% for the CBs sample. These results suggest a more uniform phase distribution and improved film homogeneity across the probed area in the CB/IPAs12 sample (see **Figures S6** and **S7**). To assess the structural differences between CBs and CB/IPAs12 samples, we performed in-situ Grazing-Incidence Wide-Angle X-ray Scattering (GIWAXS) measurements during both the perovskite soaking step (40–60 s) and the annealing treatment (60–120 s). The whole deposition protocol

is represented in **Figure S8**. As shown in **Figures 1f** and **1g**, both CB and CB/IPAs12 samples show the appearance of a peak associated with the non-perovskite δ -phase ($q=0.85 \text{ \AA}^{-1}$) during antisolvent soaking, followed by the appearance of a peak related to the cubic α -phase ($q=1.00 \text{ \AA}^{-1}$) after the beginning of the annealing process (see **Figures S9** and **S10** for further details). **Figure S11** reports the evolution of the diffraction peak FWHM related to the cubic phase for both CBs and CB/IPAs12. The narrower peak observed for the CB/IPAs12 sample indicates larger crystallite domains, according to the Scherrer equation, compared to CBs, confirming the improved crystallinity of the former sample.⁴¹ **Figure S12** shows Scanning Electron Microscopy (SEM) images of the samples processed with different solvents. The images reveal an increase in the average grain size for the CB/IPAs12 samples, which is beneficial for charge transport due to the lower density of grain boundaries.^{40,42}

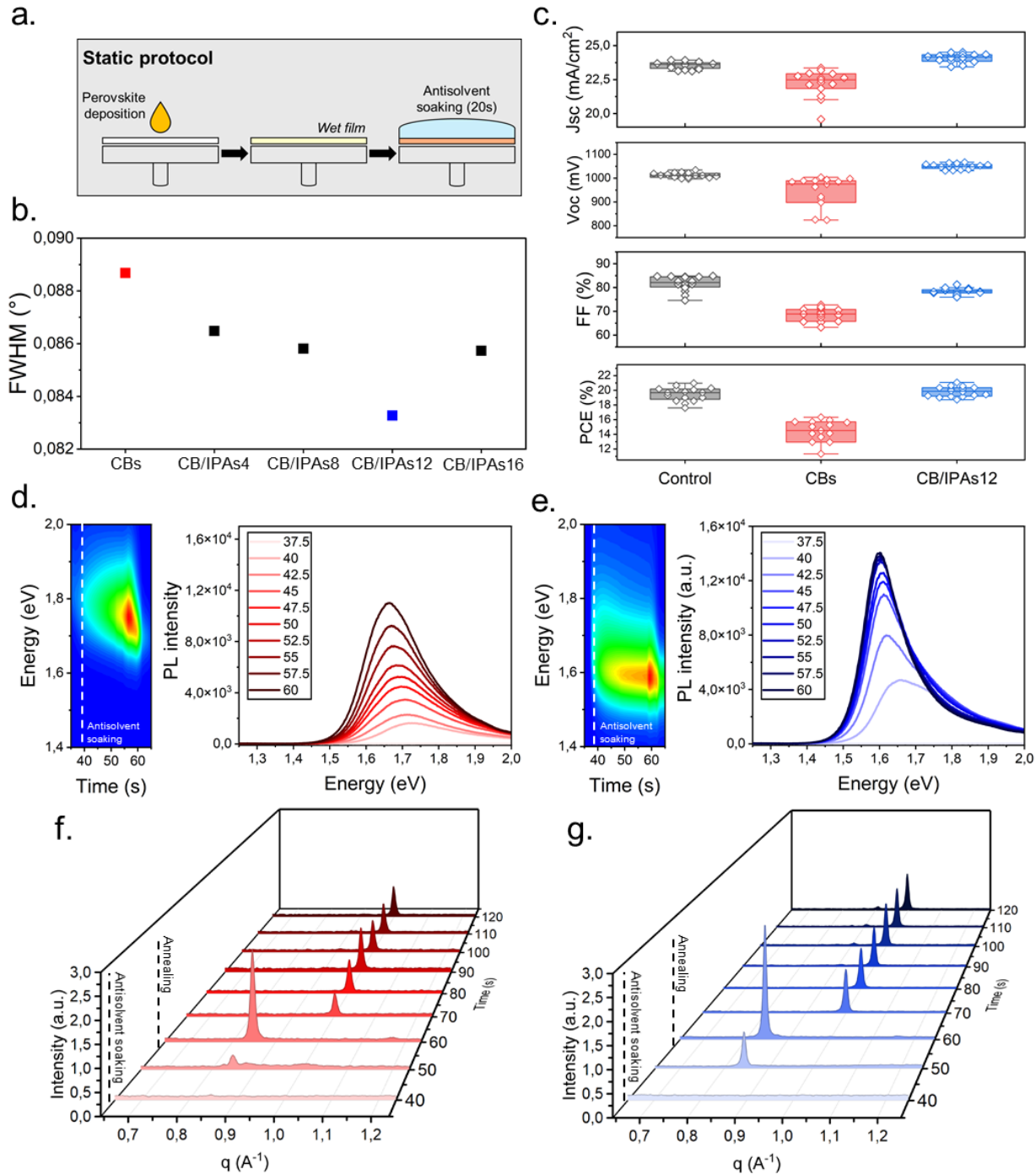


Fig. 1 a) Schematic representation of the static antisolvent procedure. b) Comparison of the FWHM of the (100) reflection signal intensity for α -phase FAPbI₃ thin films statically treated with pure CB (CBs) and different CB/IPA ratios, as obtained from X-ray diffraction patterns. c) Box charts representing J_{sc} , V_{oc} , FF and PCE of p-i-n devices dynamically treated with CB (control) and statically treated either with CB (CBs) or CB/IPA 12:1 v/v (CB/IPAs12). Results of in-situ photoluminescence (PL) measurements during the dipping step with d) of CBs and e) CB/IPAs12, highlighting different peak positions and intensities for the two samples. In-situ Grazing-Incidence Wide-Angle X-ray Scattering (GIWAXS)

measurements during both the antisolvent dipping step and the annealing treatment for f) CBs and g) CB/IPAs12 samples, showing the transition from δ -phase ($q=0.86$) to α -phase ($q=1.00$).

Validation under Humid Ambient Conditions and with Green Antisolvents

Given the beneficial effect of the static CB/IPAs12 binary solvent composition on perovskite crystallization, we attempted to transfer our static process from an inert atmosphere to ambient conditions (30% relative humidity, RH), crucial from a commercial perspective.^{43–45} To address the environmental concerns associated with aromatic and halogenated solvents, chlorobenzene (CB) was also replaced with the eco-friendlier ethyl acetate (EtAc).^{46–48} A schematic representation of the process is shown in **Figure 2a**. The PV parameter distribution and J–V characteristics (**Figures 2b–c**) reveal that, unlike devices fabricated under controlled atmosphere (**Figure 1c**), samples prepared in ambient conditions with CB/IPAs12 outperform not only CBs-based devices but also the control devices fabricated using the standard dynamic approach (see **Figure S13**). In particular, static devices processed with CB/IPAs12 and EtAc/IPAs12 achieved efficiencies of around 18%, compared to ~15% for both control and CBs devices with a net improvement of all PV parameters and a narrower statistical distribution, proving the robustness of the approach. We attribute this improvement to the more protective nature of the static process, which confines perovskite crystallization within the liquid environment and limits interaction with deleterious H₂O molecules that can lead to the formation of unwanted δ -phase (see **Figure S14**). Notably, the beneficial effect of IPA is preserved when combined with EtAc, confirming its potential as a greener alternative, as demonstrated by CB/IPAs12 and EtAc/IPAs12 devices which exhibit 20–30% higher performance compared to control and CBs samples (**Figure 2d**). Beyond enhanced efficiency and reproducibility, devices fabricated with the binary solvent system also show reduced hysteresis compared to CBs-based devices (see **Figure S15**).

It is worth mentioning that, although the use of the static protocol, especially when coupled with ethyl acetate, can represent an effective strategy to improve the environmental compatibility of the perovskite

deposition process, additional aspects such as the total amount of solvent employed and the replacement of hazardous solvents like chloroform and DMF must also be considered in order to further enhance the sustainability of perovskite processing.^{49,50}

Moreover, we investigated the use of butanol (BuOH, which is less polar than IPA) and ethanol (EtOH, which is more polar than IPA) as co-solvents in the binary antisolvent system. As shown by the box charts (**Figure S16**), butanol exhibits an effect comparable to that of IPA, leading to improved device reproducibility and enhanced power conversion efficiency. In contrast, devices fabricated using ethanol show an overall decrease in all PV parameters, including open-circuit voltage, short-circuit current density, fill factor, and efficiency. These findings indicate that the polarity of the binary solvent system plays a crucial role in determining the quality of the perovskite layer and, consequently, the performance of the resulting devices.

We next examined the samples fabricated in ambient air using time-resolved photoluminescence (TRPL) measurements (**Figure S17**). Both the PL intensity and decay dynamics clearly reflect the higher quality of the films processed with the binary solvent system, showing stronger emission and longer carrier lifetimes for CB/IPAs12 and EtAc/IPAs12 compared to CBs and control samples, indicating lower non-radiative recombination.^{51,52} We next examined the ambient air (50% of relative humidity) stability of all devices fabricated in ambient air using the static approach under maximum power point tracking (MPPT, according to the ISOS-L-1 protocol), **Figure 2e**. All the samples processed with the static approach showed an improved stability ($t_{80} \sim 200\text{h}$) compared to the control devices ($t_{80} \sim 80\text{h}$), confirming the effectiveness of our strategy in enhancing not only performance but also operational stability due to improved phase homogeneity and reduced hysteresis.⁵³ An additional advantage of using a higher fraction of polar components in the antisolvent mixture is that it enables the incorporation of long-chain organic salt that otherwise would not be soluble in a single low-polarity solvent.⁵⁴ To prove this, 4-methylphenethylammonium bromide (MPEABr) in the antisolvent mixture. **Figure S18** reports the

comparison between the CB/IPAs12 and the MPEABr-based devices; the latter shows an improved PCE, consistent with our previous work, indicating that our method is a promising strategy to achieve synergistically enhanced crystallization and defect passivation.⁵⁵

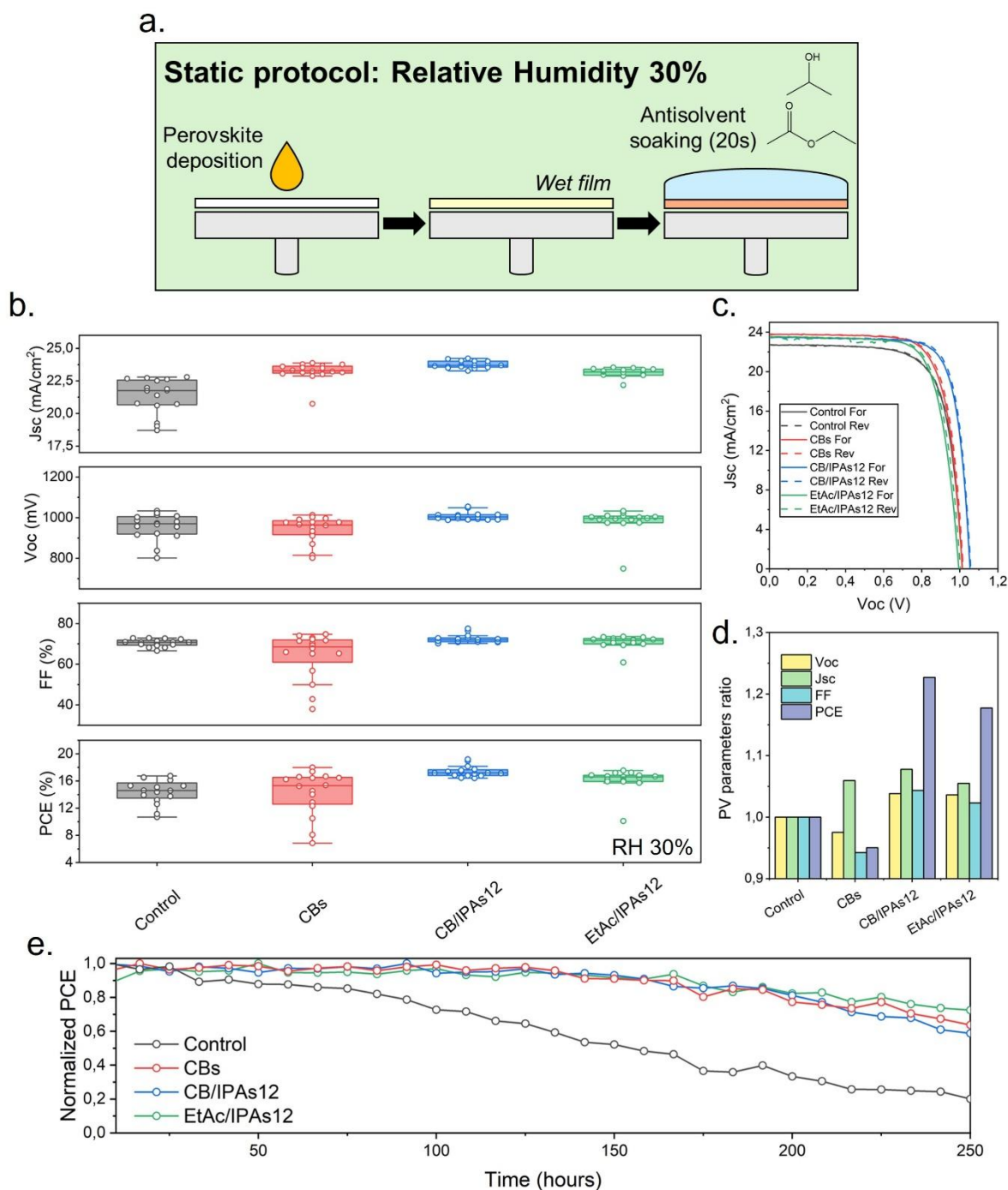


Fig. 2 a) Schematic representation of the static antisolvent procedure where CB is replaced by eco-friendly ethyl acetate and the treatment is operated in ambient conditions. b) Box charts representing the main PV

parameters and c) J-V characteristics both in forward and reverse scan for p-i-n devices based on control CBs, CB/IPAs12 and EtAc/IPAs12 samples. d) Histogram comparing the PV parameters of statically treated devices with the ones of dynamically treated (control) ones; values are reported in terms of ratio with respect to the ones exhibited by control devices. e) PCE tracking (at maximum power point) of control devices fabricated in ambient air, as compared to the one of statically treated CB/IPAs12 and EtAc/IPAs12 devices fabricated under the same conditions.

Extension to Mini-Modules and Tandem Solar Cells

To verify the broad applicability of our method, we tested it on three additional triple-cation perovskite compositions with different bandgaps: $\text{FA}_{0.9}\text{MA}_{0.07}\text{Cs}_{0.03}\text{PbI}_3$ (1.53 eV), $\text{MA}_{0.95}\text{Cs}_{0.05}\text{Pb}(\text{I}_{0.95}\text{Br}_{0.05})_3$ (1.64 eV), and $\text{MA}_{0.95}\text{Cs}_{0.05}\text{Pb}(\text{I}_{0.85}\text{Br}_{0.15})_3$. As shown by the box charts in **Figure S19**, the static method also resulted in a narrow distribution of photovoltaic parameters in these cases, demonstrating its effectiveness regardless of the perovskite composition used. We further evaluated the scalability of our method by fabricating perovskite-based minimodules with a bandgap of 1.64 eV and an active area of 25 cm². The schematic architecture of the minimodule is shown in **Figure 3a**. Devices fabricated using the static approach with the CB/IPAs12 solvent system achieved a power conversion efficiency (PCE) of 18.1% (**Figures 3c and 3e**). Importantly, the decrease in performance observed when moving from small-area devices to mini-modules is mainly due to efficiency losses caused by higher series resistance (e.g. increase of contact resistance).^{56,57} Moreover, current collection losses become more significant as the total current increases, while larger cells can also suffer from greater optical losses due to metal grid shading and dead areas, as well as more pronounced non-uniformities in current distribution across the device, all of which contribute to lower overall efficiency.^{58–60} To further demonstrate the versatility of the strategy and surpass the efficiency limits of single-junction photovoltaics, we subsequently implemented the method in a two-terminal monolithic perovskite/silicon tandem configuration (**Figure 3b**).⁶¹ A highly effective strategy widely adopted in the perovskite community is the integration of a wide-bandgap perovskite absorber on top of a silicon bottom cell. In this architecture, careful tuning of the perovskite bandgap is essential to

achieve current matching between the sub-cells and thus maximize the overall power output.⁶² For this reason, we used a perovskite composition with a target bandgap of 1.64 eV, suitable for ensuring current matching in monolithic perovskite/silicon tandems. The resulting 1 cm² monolithic perovskite/silicon tandem devices (**Figure 3b**) delivered a power conversion efficiency of 27.68% (**Figures 3d and 3e**).

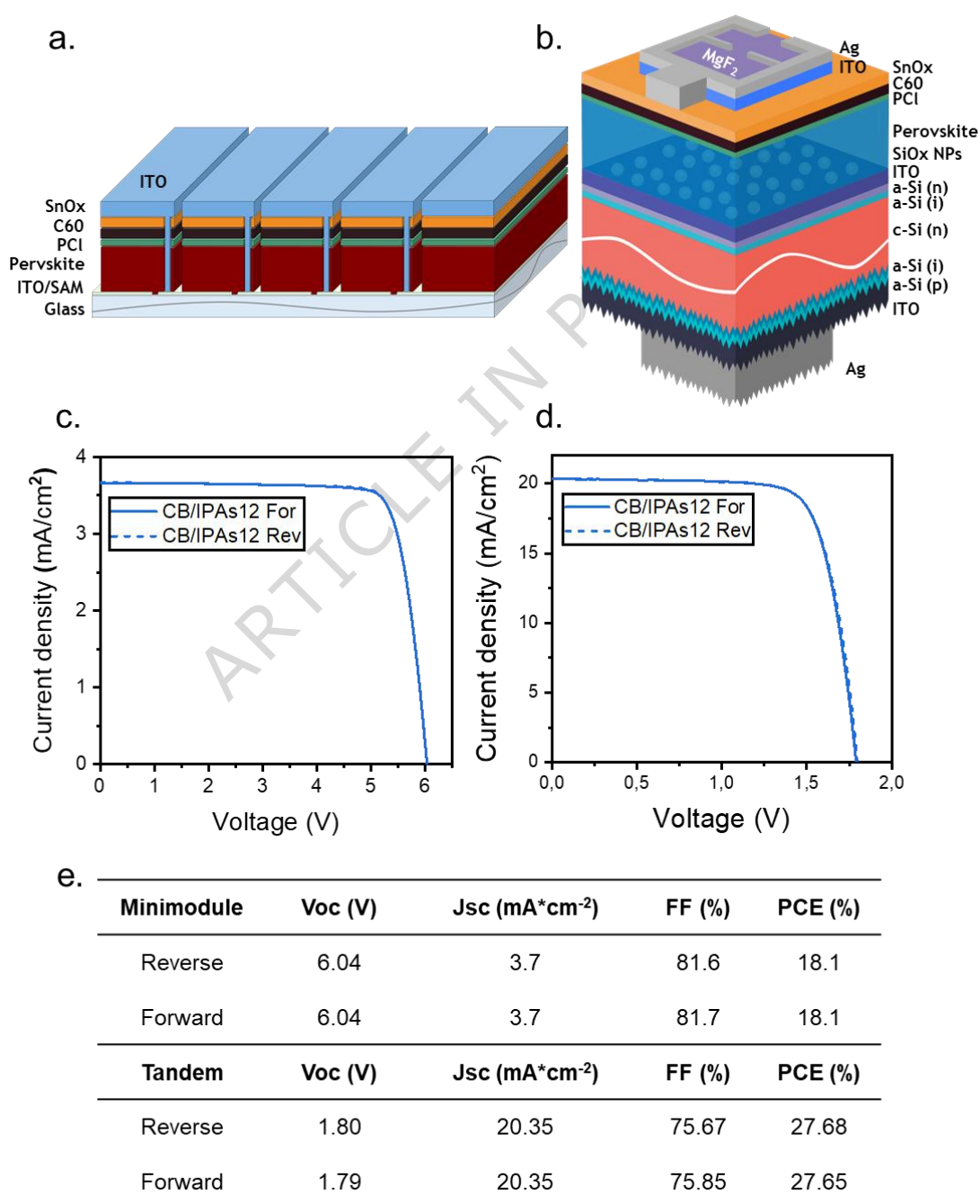


Fig. 3 Schematic representation of a) single-junction minimodules and b) tandem devices, with an active area of 25 cm² and 1 cm², respectively, that were employed to test the static antisolvent treatment. J-V

characteristics in both forward and reverse scan for the best c) single-junction minimodule and d) tandem device in which the static procedure was used. e) Table reporting the PV parameters of minimodule and tandem devices processed with the static method.

Conclusions

This work introduces a generalizable static solvent treatment strategy that enables the formation of homogeneous perovskite layers through controlled post-deposition crystallization. By combining a polar solvent, isopropanol, with low-polarity antisolvents such as chlorobenzene or the “greener” alternative ethyl acetate, the method promotes uniform crystallization of the perovskite film. In-situ photoluminescence and in-situ GIWAXS showed that the polar component plays a key role in accelerating perovskite crystallization and promoting the formation of larger crystalline domains, ultimately leading to a homogeneous perovskite layer. Unlike conventional dynamic antisolvent approaches, which often require extensive optimization depending on perovskite composition, film thickness, and device architecture, the static treatment developed here shows broad applicability and reduced sensitivity to processing parameters. In particular, devices processed in humid environment (RH=30%) with the static method showed performances comparable with devices processed with the dynamic approach in a controlled atmosphere. This versatility is demonstrated across multiple perovskite compositions spanning different bandgaps and on 25 cm² perovskite minimodules and monolithic perovskite/silicon tandem devices, achieving power conversion efficiencies of 18.1% and 27.7%, respectively. Importantly, the static nature of the process improves reproducibility and simplifies scale-up, two key requirements for industrial translation. The compatibility with environmentally friendlier solvents such as ethyl acetate further enhances the sustainability of the process, addressing concerns associated with commonly used aromatic or halogenated antisolvents. Overall, this strategy provides a practical pathway toward scalable, reproducible and environmentally compatible manufacturing of next-generation perovskite photovoltaic technologies.

Methods

Materials

Chlorobenzene (CB, extra dry, 99.8%), dimethyl sulfoxide (DMSO, $\geq 99.9\%$ extra dry), N,N-dimethylformamide (DMF, extra dry, 99.8%), ethyl acetate (EtAc, extra dry, 99.8%), and chloroform (CF, extra dry 99.8%) were purchased from Acros Organics. 2-propanol (IPA, $\geq 99.8\%$) lead iodide (PbI_2 , $> 98.0\%$) and MeO-2PACz were purchased from TCI. Formamidinium iodide (FAI, $> 99.99\%$), methylammonium bromide (MABr, $> 99.99\%$) cesium iodide (CsI, $> 99.99\%$) and methylammonium chloride (MACl, $> 99.99\%$) were purchased by Greatcell. Phenyl-C61-butyric acid methyl ester (PCBM, $> 99.99\%$) was purchased from Lumatec. Ethanol (extra dry, 99.8%) and bathocuproine (BCP) were purchased from Sigma Aldrich. All solutions were prepared in an Ar-filled glovebox.

Perovskite solution preparation

The perovskite composition with a bandgap of 1.53 eV has been prepared by dissolving 272.5 mg FAI, and 774.5 mg of PbI_2 and 37.8 mg of MACl in 1 mL DMF/DMSO 4/1 vol/vol solution.⁶³ The triple cation perovskite composition with a bandgap of 1.53 eV has been prepared by dissolving 247.6 mg FAI, 20.34 mg of MAI, 20.8 mg of Cs and 774.5 mg of PbI_2 in 1 mL DMF/DMSO 4/1 vol/vol solution. The perovskite composition with a bandgap of 1.64 eV has been prepared by dissolving 226.5 mg of MAI, 19.5 mg of CsI, 639.7 mg of PbI_2 and 41.3 mg of PbBr_2 in 1 mL DMF/DMSO 4/1 vol/vol solution. The perovskite composition with a bandgap of 1.68 eV has been prepared by dissolving 226.5 mg of MAI, 19.5 mg of CsI, 535.9 mg of PbI_2 and 123.9 mg of PbBr_2 in 1 mL DMF/DMSO 4/1 vol/vol solution.

Single-junction device fabrication (dynamic method)

For the fabrication of PSCs, indium tin oxide (ITO)-coated glass substrates (purchased from Yingkou Shangneng Photoelectric Material Co.,Ltd.) were consecutively cleaned in acetone and IPA by ultrasonicing for 15 min for each solvent. Substrates were dried with N₂ airflow and O₂ plasma treated for 10 min. MeO-2PACz was dissolved in ethanol and 50 μL were spin-coated onto ITO/glass substrates at 3000 rpm for 30 s and annealed at 100 °C for 10 min. 25 μL of the final solution were deposited on the MeO-2PACz coated substrates and spin-coated with a three-step procedure: the first step proceeded at 1000 rpm (500 rpm/s) for 6 s, the second step proceeded at 5000 rpm for 27 s, while the last step was a speed reduction of 1250 rpm/s. 200 μL of CB, was dropped onto the spinning substrate for an antisolvent procedure after 20 seconds from the beginning of the second step. Subsequently, substrates were annealed at 100 °C for 30 min. To fabricate the ETL, PCBM was dissolved in chloroform to produce a 15 mg mL⁻¹ solution. 20 μL of the solution were spin-coated at 2000 rpm for 20s onto the perovskite layer. To prevent the diffusion of the metal contact inside the perovskite 50 μL of 0.5 mg mL⁻¹ of bathocuproine solution (in isopropanol) were deposited on PCBM. Finally, 80 nm Ag was thermally evaporated onto the device with a shadow mask with an area of 0.0825 cm². The evaporation speed was adjusted to 0.01 nm s⁻¹ for the first 5 nm, 0.02 nm s⁻¹ from 5 to 15 nm, and 0.06 nm s⁻¹ for the rest of the procedure. A vacuum-based chuck was used throughout the entire deposition process. The deposition of each layer of the solar cell was performed in an N₂-filled glovebox.

Single-junction device fabrication (static method)

For the fabrication of PSCs, indium tin oxide (ITO)-coated glass substrates (purchased from Yingkou Shangneng Photoelectric Material Co.,Ltd, 1.2x1.2 cm²) were consecutively cleaned in acetone and IPA by ultrasonicing for 15 min for each solvent. Substrates were dried with N₂ airflow and O₂ plasma treated for 10 min. MeO-2PACz was dissolved in ethanol and 50 μL were spin-coated onto ITO/glass

substrates at 3000 rpm for 30 s and annealed at 100 °C for 10 min. 25 μL of the perovskite solutions (with different compositions as described in the “perovskite solution preparation” section) were deposited on the MeO-2PACz coated substrates and spin-coated with a three-step procedure: the first step proceeded at 1000 rpm (500 rpm/s) for 6 s, the second step proceeded at 5000 rpm for 27 s, while the last step was a speed reduction of 1250 rpm/s. The resulting wet-perovskite layer was dipped into a petri-dish containing 15 mL of antisolvent (CB, CB/IPA12, CB/EtOH12, CB/BuOH12, EtAc or EtAc/IPA12). Subsequently, substrates were annealed at 100 °C for 30 min. To fabricate the ETL, PCBM was dissolved in chloroform to produce a 15 mg mL^{-1} solution. 20 μL of the solution were spin-coated at 2000 rpm for 20s onto the perovskite layer. To prevent the diffusion of the metal contact inside the perovskite 50 μL of 0.5 mg mL^{-1} of bathocuproine solution (in isopropanol) were deposited on PCBM. Finally, 80 nm Ag was thermally evaporated onto the device with a shadow mask with an active area of 0.0825 cm^2 . The evaporation speed was adjusted to 0.01 nm s^{-1} for the first 5 nm, 0.02 nm s^{-1} from 5 to 15 nm, and 0.06 nm s^{-1} for the rest of the procedure. A vacuum-based chuck was used throughout the entire deposition process. The deposition of each layer of the solar cell was performed either in a nitrogen atmosphere (for the devices shown in Figure 1c) or in a dry box under 30% relative humidity at room temperature (for the devices shown in Figure 2).

Tandem device fabrication

The bottom cell fabrication process is performed on 210- μm -thick n-type, FZ Si wafers. Wafers are textured using alkaline solutions to create a random pyramid pattern, then wafers undergo a cleaning sequence prior thw PECVD process. Through PECVD (Octopus II, Indeotec), hydrogenated amorphous Si layers are deposited on both sides. On the front side, a-Si:H(i) and a-Si:H(n) are deposited at 200 °C. On the rear side, a-Si:H(i) and a-Si:H(p) are deposited at 200 °C. On the rear side, a 150-nm-thick NSCOT layer is deposited by sputtering through a shadow mask and then capped with 500 nm of Ag. The front side is completed by sputtering a 20-nm-thick ITO layer for interconnection. As the final step before top

cell processing, 4" wafers are then laser-cut into $2.5 \times 2.5 \text{ cm}^2$ substrates, which are annealed at 210°C for 10 min in ambient to optimize the properties of the solar cells.

To finalize the tandem device, the perovskite sub-cell was deposited on top of the silicon bottom cells following the procedure for the single junction, but with a bandgap of 1.65 eV, adjusted for tandem integration. The surface of the perovskite was passivated with piperazinium chloride deposited via spin coating (0.1 mg/ml in IPA, 6000 rpm). Then, 20 nm of evaporated C60 (Angstrom Engineering) and 15 nm of SnOx via ALD (Picosun) are deposited on top of the perovskite, followed by 40 nm of sputtered IZrO as front transparent electrode. Finally, 500 nm of Ag are evaporated to secure the contact point and 100 nm of evaporated MgF₂ acts as anti-reflective coating.

Minimodule fabrication

Minimodules were fabricated with the same procedure as the single junction devices but with the bandgap used for tandem devices (1.65 eV). The surface of the perovskite was passivated with piperazinium chloride deposited via spin coating (0.1 mg/ml in IPA, 6000 rpm). Then, 20 nm of evaporated C60 (Angstrom Engineering) and 15 nm of SnOx via ALD (Picosun) are deposited on top of the perovskite, followed by 500 nm of Ag as opaque electrode. For the minimodule integration, conventional laser patterning (p1, p2, and p3) was performed with a green pulsed picosecond laser (Trumpf).

In-situ Photoluminescence

In-situ PL measurements were taken in a nitrogen glovebox using a custom-built setup previously described in other works.⁶⁴ The setup consisted of a 405 nm laser diode, a plano-convex lens positioned above the substrate, a 450 nm long-pass optical filter, and an Ocean Optics QEPro spectrometer. The maximum power density for the in-situ PL measurements was approximately 100 mW/cm^2 , with spectra recorded continuously at 100 ms intervals. The PL emission spectra were fitted with a single Gaussian

peak after the spectrometer background was accounted for, and the data were transformed to an energy scale using Jacobian transformation.

In-situ Grazing Incidence Wide Angle X-Ray Scattering

The in-situ GIWAXS measurements were carried out at the 12.3.2 microdiffraction beamline at the Advanced Light Source (ALS) - a custom-designed analytical chamber enabled processing of the perovskite film during simultaneous GIWAXS measurements - details can be found in ⁶⁵. The glass/ITO substrates were placed onto an integrated spin coater and puck heater and secured using heat transfer paste. The analytical chamber was sealed off from the external environment by being held under nitrogen flow before the precursor solution was applied to the substrate. The X-ray incident angle for GIWAXS measurements was 1° with a beam energy of 10 keV. The sample detector distance (SDD) was ~ 155 mm, and the detector was positioned at 35° to the sample plane. The GIWAXS data were continuously recorded with a 1-second integration time using a 2D Pilatus 1 M detector (Dectris Ltd.). After closing the beamline hutch and starting the measurement, the spin coating was remotely initialized, including a remote-controlled anti-solvent dispense. At the end of the spin-coating protocol, the puck was heated to the respective annealing temperature while continuously spinning at 200 rpm to ensure uniform measurement conditions during the annealing step. The resulting GIWAXS patterns were calibrated using both an Al_2O_3 reference sample for external calibration and the ITO substrates for internal calibration of each individual sample. Data analysis was done using the Python package mmanalysis developed at BL 12.3.2.

Time-Resolved photoluminescence steady-state photoluminescence

Time-resolved photoluminescence (TRPL) curves were collected using a customized optical setup in which the excitation was provided by pulsed/CW laser at 470 nm (PicoQuant) with a spot area of $\approx 70 \mu\text{m}^2$. The luminescence was analyzed with an interferometer (GEMINI by Nireos) and recorded with a

single photon detector (IDQuantique) coupled with a Time Tagger (Swabian Instrument). The scan was performed in pulsed mode (repetition rate of 5 MHz) using an energy fluence of $0.2 \mu\text{J cm}^{-2}$.

X-Ray Diffraction

XRD patterns were acquired using a Bruker D2 X-ray diffractometer coupled with a Cu X-ray tube ($\lambda = 1.5418$, 300 W, 30 kV) and a Lynxeye (1D mode) detector. Measurements were performed adopting 0.02° step size and 1 step per s scan rate.

Scanning Electron Microscopy

Scanning electron microscopy (SEM) images of the films at different magnifications were collected using a FEI® Helios Nanolab 450S equipment using 5 kV and 0.2 nA as conditions for imaging.

Device characterization

Current density-voltage (J-V) characteristics of the produced PSCs were carried out using a Keithley 2400 source meter and an Abet solar simulator. The simulated 1 Sun AM1.5G illumination was calibrated using a certified Si reference cell (Open RR-1002, KG5 window). The devices were measured in ambient air (20°C and 50% RH) both in reverse scan ($1.2 \text{ V} \rightarrow 0 \text{ V}$, 130 mVs^{-1}) and forward scan ($0 \rightarrow 1.2 \text{ V}$, 130 mVs^{-1}). During every measurement, a metal shadow mask with an aperture area of 0.03 cm^2 was used to cover the device's active area. Incident Photon to Current Conversion Efficiency (IPCE) was performed using Cicci Research Arkeo transient tests module. Measurements were conducted under 1 Sun illumination and constant bias was perturbed with short light pulses of $200 \mu\text{s}$ width. Tandems and minimodules were characterized by a dual lamp solar simulator (Wacom) calibrated with a reference cell with dual filters. Current density-voltage (J-V) characteristics of the tandems and minimodules were carried out using a Keithley 2400 source meter; the tandems were measured both in reverse scan (1.9 V

→ -0.1 V, 200 mVs⁻¹) and forward scan (-0.1 → 1.9 V, 200 mVs⁻¹). During every measurement, a rubber shadow mask with an aperture area of 1 cm² was used to define the device's active area. Current density-voltage (J-V) characteristics of the minimodules were measured both in reverse scan (6.5 V → -0.1 V, 500 mVs⁻¹) and forward scan (-0.1 → 6.5 V, 500 mVs⁻¹).

Statistical analysis. For photovoltaic parameters statistical analysis, data were pre-processed by defining as outliers all values lying above or below the 1.5 interquartile range. The resulting dataset was arranged in box charts, where the upper and lower limits of the box identify the 25th and 75th percentile, respectively, while whiskers indicate the 1.5 interquartile range. The mean value for each sample is presented as a horizontal line within the box. OriginPro 2018 was employed for statistical analysis.

Maximum Power Point Tracking

The stability measurements were performed on unencapsulated devices at the maximum power point (according to the ISOS-L-1) under ambient air using a stability box (Litos Lite) from Fluxim, coupled with a Wavelabs solar simulator continuously illuminating the devices in ambient air (50% RH, and at a temperature ranging from 20°C to 50°C due to the additional heat generated by continuous illumination.). The spectrum used for stability measurements was the same as the one employed for current density voltage measurements.

Photoluminescence mapping

Photoluminescence mapping images were acquired using a Photon IMA-VIS hyperspectral microscope. Measurements were performed over the spectral range of 600–1000 nm using a 532 nm excitation laser with an exposure time of 1 s. A dark frame, obtained by averaging several dark images acquired under the same exposure time and detector settings as the sample measurements, was used to subtract thermally generated charges. The intrinsic spatial wavelength-dependent gradient was corrected prior to analysis.

Spatial translation and dilation between images in the wavelength-dependent sequence were corrected using the registration tool available in PhySpecV2 software.

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Author Contributions

Conceptualization: RP and GG. Methodology: RP, SF, FT, TK, MD, MDB, SM, CSF, QJ. Supervision: BMG, CSF, JQ, AO, CJB, LHE, MDB, GG. Writing: RP. Review and editing: RP, SF, FT, TK, SM, AO, BMG, CJB, CSF, GG.

Data availability

All data supporting the findings of this study can be found in the article and its Supplementary Information. More detailed primary experimental data are available from the corresponding author upon reasonable request.